

UK Visa Application Guidance & Document Readiness Agent

Evaluation Report

1. Evaluation Objective

The purpose of the evaluation was to determine whether the UK Visa Guidance Agent behaves reliably across different types of user interactions.

The evaluation was not limited to checking whether the model could produce a reasonable-looking answer. The project was designed to test the complete agent workflow, including retrieval, tool selection, planning, memory, adaptive behaviour and failure handling.

The evaluation therefore considered both the quality of the final answer and the behaviour that produced it.

2. Evaluation Method

A set of representative test scenarios was created around the main capabilities of the agent.

The test set included:

- Visa document questions
- Skilled Worker questions
- Student visa questions
- Visitor visa questions
- Basic eligibility questions
- Multi-step questions
- Follow-up questions
- Questions requiring memory
- Questions requiring tool selection
- Questions where relevant information is missing
- Out-of-domain questions
- Invalid inputs
- Runtime/API failure scenarios

Each test was assessed against criteria such as:

Accuracy, relevance, completeness, groundedness, hallucination, tool selection and expected behaviour.

Repeated questions were also used to assess consistency.

The detailed results are recorded in the project's **Evaluation Results** spreadsheet. The final submission should use the actual figures from that evaluation rather than estimated scores.

3. Key Evaluation Scenarios

Scenario 1 — RAG Retrieval

Question:

What documents are required for a Skilled Worker visa?

The expected behaviour was for the agent to identify this as a visa guidance question, search the vector database and use relevant GOV.UK content when generating the answer.

This test evaluates whether the RAG pipeline is actually connected to the final response rather than simply being present in the architecture.

Scenario 2 — Multi-Step Planning

Question:

I have a sponsor, passport and English qualification. Am I eligible and what documents do I need?

This requires more than a single retrieval operation.

The expected flow is:

Question → Planning → Eligibility check → Document retrieval → Response

This test was used to demonstrate that the system can break a request into separate tasks and use the appropriate tools.

The eligibility result is intentionally treated as a basic preliminary check rather than a guarantee that the applicant qualifies for a visa.

Scenario 3 — Conversation Memory

The user first asks:

What documents are required for a Skilled Worker visa?

The user then asks:

What about my wife?

The second question is deliberately incomplete on its own. The expected behaviour is that the agent uses the previous conversation context to understand the reference to the user's wife in relation to the Skilled Worker visa.

This evaluates the short-term memory functionality introduced during the project.

Scenario 4 — Adaptive Behaviour

The user provides the feedback:

Please answer using short bullet points.

The user then asks another visa question.

The expected behaviour is that the response follows the requested format.

This evaluates whether feedback is stored and whether it actually changes future agent behaviour.

Scenario 5 — Missing or Unsupported Information

The user asks:

Tell me about Mars visas.

The expected behaviour is not to invent a fictional visa category. Instead, the agent should indicate that it cannot find relevant information in its knowledge base.

This is an important safety test because an answer can appear fluent while still being completely unsupported.

Scenario 6 — Out-of-Domain Question

The user asks:

What is the weather in London today?

The system should recognise that this is outside the intended UK visa guidance domain rather than pretending that it has the appropriate information or tool.

This tests the domain boundary and tool-selection safeguards.

4. Results and Observations

The evaluation demonstrated that the agent can combine retrieval with LLM generation and can use the additional capabilities introduced during the later phases of development.

The RAG component provides a way of grounding answers in the project's selected GOV.UK documents. This is important because a general-purpose LLM may otherwise rely on information from its training data that could be incomplete or outdated.

The tool-use tests also demonstrated the importance of routing. The agent should not use every tool for every question. A document-related question should trigger document retrieval, while an unrelated question should not result in an inappropriate tool call.

Memory improved the quality of multi-turn conversations because users could refer back to information already established in the conversation.

The adaptive behaviour tests showed that user feedback can be used to influence later responses, making the interaction more personalised without changing the underlying knowledge base.

Actual evaluation metrics:

Insert the final accuracy, pass rate, consistency, or other measured values from the Evaluation Results tab here.

5. Failure and Root Cause Analysis

One important failure mode is **retrieval failure**. If the relevant document or information has not been indexed, the system cannot retrieve it reliably.

The root cause may therefore be document coverage, chunking or retrieval quality rather than the LLM itself.

A second failure mode is **outdated information**. GOV.UK guidance can change, but the vector database only knows about the material that has been indexed. This makes document-refresh automation an important future requirement.

Another failure mode is **API quota exhaustion**. During development, Gemini API quota limits were encountered. This demonstrated that an otherwise functioning agent can still fail because of external service constraints. Retry handling, rate-limit awareness and graceful failure are therefore necessary for deployment.

There is also a risk of ambiguous questions. If the user has not provided enough information, the agent should ask for clarification rather than confidently making assumptions.

6. Safety and Ethics Review

The most important safety concern is that a user could interpret the agent's response as definitive immigration or legal advice.

The agent therefore needs to clearly communicate its limitations. It should provide guidance based on the available official documents but should not guarantee a visa outcome.

Other risks include:

- Hallucinated information
- Outdated guidance
- Incorrect interpretation
- Privacy concerns

- Over-reliance on automated answers

Using official GOV.UK material for retrieval reduces the risk of unsupported answers, although it cannot remove the risk completely.

7. Improvements Identified

The evaluation highlighted several areas for future improvement.

The highest-priority improvement is automatic monitoring of GOV.UK guidance. When official documents change, the knowledge base should be refreshed and embeddings regenerated.

Other improvements include:

- Better retrieval-quality testing
- Larger evaluation datasets
- More comprehensive eligibility rules
- Improved ambiguity detection
- Stronger monitoring and tracing
- Better API retry and fallback behaviour
- More extensive safety testing

8. Conclusion

Overall, the evaluation shows that the project has progressed beyond a basic chatbot. The agent can retrieve information, use tools, maintain conversation context, adapt to feedback and handle several failure scenarios.

The evaluation also demonstrates that the system is still a prototype. The main limitations are knowledge freshness, retrieval coverage, API dependency and the complexity of real UK immigration rules.

For this reason, the appropriate product position is a **UK visa guidance and document-readiness assistant**, rather than an automated immigration decision-maker.